A Formula-by-Formula Guide to
Causal Inference for Group-Contaminated Structured Outcomes

Explanatory companion to the paper by Usef Faghihi and Amir Saki

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1 What the paper is about

The paper studies causal effects when an outcome is a structured object—for example, a multichannel microscopy image—but the recorded object has first been moved, rotated, or otherwise transformed in an unknown, unit-specific way. The difficult case is allowed: the transformation may depend on treatment, covariates, or the intrinsic biological outcome. Therefore, a difference between two raw images can be caused either by biology or by acquisition geometry.

The paper asks three different questions that must not be conflated.

1. **Observability:** What information about the intrinsic outcome survives every allowed transformation? The answer is exactly the information that is constant on group orbits.
2. **Causal identification:** Under what treatment-design assumptions can the distribution of that surviving information under treatment be recovered? The answer uses quotient consistency, conditional exchangeability, and positivity.
3. **Statistical losslessness:** When does replacing the raw observation by its quotient lose no information about the whole statistical model? Maximal invariance alone is not enough. A parameter-free conditional law inside each orbit is necessary and sufficient; conditional Haar contamination is one special case that supplies it.

1.1 What "canonical image" means

A *canonical image* is one deterministically selected representative from all translated and rotated versions of the same image.

It is not necessarily the original image, the biologically "correct" orientation, or an estimated clean image.

1. The orbit contains all equivalent images

For an image y, its orbit is

$$[y]_G = \{g \cdot y : g \in G\}.$$

Read as: Start with y and apply every allowed transformation g. The resulting set contains all images considered equivalent to y.

For example, if G contains integer translations and rotations by $0°$, $90°$, $180°$, and $270°$, the orbit contains every shifted and quarter-turned copy of y.

2. Selecting one standard representative

The canonicalization function c selects exactly one standard image from this orbit:

$$c(y) \in [y]_G.$$

Read as: The canonical image $c(y)$ is one member of the orbit of y, chosen by a fixed deterministic rule.

For the paper's finite-support lattice images, the construction is as follows.

1. Move the lower corner of the nonzero bounding box to the origin.
2. Construct the four quarter-turn versions:

   $$x_r = N\{(0, r) \cdot x\}, \qquad r \in C_4 = \{0, 1, 2, 3\}.$$

   Read as: For each allowed quarter turn r, rotate x and then use N to translate the rotated image back to the standard origin.

3. Serialize the pixels of every version using the fixed key κ.
4. Select the lexicographically smallest version:

$$c(x) = x_{r^\star}, \qquad r^\star = \min \arg\min_{r \in C_4} \kappa(x_r).$$

Read as: Among the four translated-to-the-origin orientations, choose the one whose pixel representation comes first under a fixed ordering rule; if several orientations tie, choose the smallest rotation index.

3. A simple pixel example

Suppose an image contains this L-shape, and its four normalized rotations are

$$y = x_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad x_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Read as: The four matrices show the same three-pixel L-shape in its four quarter-turn orientations after each orientation has been moved to the standard origin.

Using row-by-row serialization, their pixel keys are

$$\kappa(x_0) = (1,0,1,1), \quad \kappa(x_1) = (0,1,1,1),$$
$$\kappa(x_2) = (1,1,0,1), \quad \kappa(x_3) = (1,1,1,0).$$

Read as: Each key lists a matrix's pixels row by row so that the four orientations can be compared using one fixed lexicographic order.

The lexicographically smallest key is

$$(0,1,1,1),$$

Read as: This key comes first because it begins with 0, whereas the other three keys begin with 1.

Therefore,

$$c(y) = x_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Read as: The matrix x_1 is the canonical image selected for the entire translation-and-rotation orbit of this L-shape.

If the observed image is translated and rotated,

$$X = g \cdot y,$$

Read as: The recorded image X is some allowed moved version of the intrinsic image y.

It still gives the same canonical image:

$$c(X) = c(g \cdot y) = c(y).$$

Read as: Canonicalizing any translated or rotated copy of y returns the same selected representative.

4. What has and has not been recovered?

Suppose

$$X = \Gamma \cdot Y(A).$$

Read as: The observed image is the intrinsic potential outcome under the received treatment after the unknown acquisition transformation Γ.

Canonicalization gives

$$c(X) = c\{\Gamma \cdot Y(A)\} = c\{Y(A)\}.$$

Read as: The observed and intrinsic images produce the same canonical representative because they differ only by an allowed nuisance transformation.

However,

$$c(X) \neq Y(A) \quad \text{may still hold.}$$

Read as: The canonical representative need not equal the intrinsic image in its original position and orientation.

For example, the intrinsic image and its canonical representative might be

$$Y(A) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad c\{Y(A)\} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Read as: The intrinsic L-shape can begin in one orientation while the deterministic ordering rule selects a different orientation as canonical.

Therefore, canonicalization does not recover the original orientation. It preserves the structure after position and orientation have been declared irrelevant.

5. Relationship with the maximal invariant

The canonicalizer satisfies

$$c(y) = c(z) \quad \Longleftrightarrow \quad z = g \cdot y \text{ for some } g \in G.$$

Read as: Two images have the same canonical image exactly when one is an allowed transformed version of the other.

This makes c a maximal invariant. In the lattice-image implementation, the abstract map M is represented by this canonicalizer:

$$M(y) = c(y).$$

Read as: In the concrete image algorithm, computing the abstract orbit label $M(y)$ means computing the deterministic canonical representative $c(y)$.

Thus, when the guide later refers to the law of canonical images, it means

$$\text{Law}\{Q(a)\} = \text{Law}\big[c\{Y(a)\}\big].$$

Read as: The treatment-specific quotient law is the probability distribution of the canonical representative selected from each intrinsic image's orbit.

A precise wording used later in F19 is: "From the identified distribution of canonical representatives—one deterministic image for each translation-and-rotation orbit—apply $\bar{h}$, such as pixel count, to obtain the treatment-specific distribution of the invariant target."

The theory is then turned into an algorithm for finite-support lattice images: remove integer translations, compare four quarter-turn orientations, choose one canonical representative, compare quotient distributions with a characteristic Gaussian kernel, and obtain a finite-sample paired randomization test by enumerating all label swaps.

1.2 The problem in one concrete example

Suppose treatment changes a cell from a three-pixel L-shape to a four-pixel square. The microscope may translate the image and rotate it by 0°, 90°, 180°, or 270°. Worse, treated images might systematically receive different rotations from control images. A raw-pixel test can then report an effect even when biology did not change.

Let G be the allowed motion group. All translated and quarter-turned versions of one intrinsic image form an *orbit*. Pixel count and shape up to motion are orbit-invariant. Absolute left-right position is not. The paper's main causal object is the orbit, encoded by a maximal invariant M, not the unknown original coordinate frame.

Throughout this guide, the running example is:

- $A \in \{0, 1\}$ is control or treatment;
- $Y(0)$ is a three-pixel L-shape and $Y(1)$ is a four-pixel square;
- Γ is an unknown integer translation followed by a quarter turn;
- $X = \Gamma \cdot Y(A)$ is the recorded image;
- $M(X)$ is a deterministic canonical version of the image;
- C is a pretreatment batch or cell-type covariate.

1.3 The complete logical chain

Layer	Mathematical question	What establishes it
Observation	Can a target be decoded for every unknown transformation?	Orbit invariance; maximal invariants; Borel factorization.
Causality	Can the treatment-specific quotient law be learned?	Consistency, conditional exchangeability, positivity, adjustment.
Information	Is quotient reduction as informative as raw data for every parameter?	A common quotient-faithful reconstruction kernel; Haar is a sufficient special case.
Robustness	What happens under imperfect transformations?	Orbit distance, a Lipschitz quotient map, Wasserstein and MMD bounds.
Algorithm	How is this done for lattice microscopy?	Translation normalization, finite rotation minimization, Gaussian quotient kernel.
Inference	How is a valid p-value obtained?	Complete enumeration of the paired assignment mechanism under a sharp null.

2 Notation before reading the formulas

Symbol	Meaning
A, a	Realized treatment and a possible treatment value.
$Y(a)$	Intrinsic potential structured outcome if treatment were set to a.
Γ, g	Unobserved realized transformation and a generic allowed transformation.
G	Transformation group.
$g \cdot y$	Result of applying transformation g to structured object y.
X	Recorded, transformed outcome.

C	Observed pretreatment covariates.
$[y]_G$	Orbit of y: all objects obtainable from y by transformations in G.
M	Maximal invariant: a complete label of an orbit.
$Q(a)$	Quotient potential outcome $M\{Y(a)\}$.
O_{full}	Full observed datum (C, A, X).
O_{quot}	Reduced datum $(C, A, M(X))$.
θ	Parameter indexing a family of observed-data laws.
K	Markov reconstruction kernel.
μ_G	Normalized Haar probability measure on a compact group.
d_G	Distance between orbits after optimizing over a group transformation.
W_1	Wasserstein-1 distance between probability laws.
$k, \Phi, \mathcal{H}_k$	Kernel, its feature map, and its reproducing-kernel Hilbert space.
MMD_k	Maximum mean discrepancy between two probability laws.
B	Number of paired randomization blocks; in Appendix A, it also denotes a generic Borel set, as stated locally.

3 Formula-by-formula explanation

3.1 Section 1: observation model

F1. The observation equation (paper Eq. 1.1)

$$X = \Gamma \cdot Y(A).$$

Read as: The recorded outcome X is obtained by taking the intrinsic outcome corresponding to the treatment actually received, $Y(A)$, and applying the unknown transformation Γ.

Simple example. If $A = 1$, $Y(1)$ is a square, and Γ means "rotate 90° and shift five pixels right," then X is that moved square.

Relationship and why the paper needs it. This is the source of the entire problem. Because Γ is unrestricted, the raw coordinate system cannot be trusted. Every later formula either characterizes what survives this equation, identifies its treatment-specific law, or builds a valid test from it.

3.2 Section 2.1: measurable group contamination and sharp observability

F2. A measurable group action

$$(g, y) \longmapsto g \cdot y.$$

Read as: Give the rule both a transformation g and an object y; it returns the transformed object $g \cdot y$.

Simple example. A pair $g = (t, r)$ may mean translate by $t = (5, -2)$ and rotate by $r = 1$ quarter turn. Applying it to an image produces a moved image.

Relationship and why the paper needs it. Formula F1 uses this action. Joint Borel measurability is needed so that transformed outcomes, conditional laws, orbit kernels, and all later probability statements are mathematically well-defined.

F3. Orbit of an outcome

$$[y]_G = \{g \cdot y : g \in G\}.$$

Read as: The orbit of y is the set of every version of y obtainable using an allowed transformation.

Simple example. If G contains four quarter turns, the orbit of an asymmetric L-shape contains its four orientations; adding translations includes every shifted copy too.

Relationship and why the paper needs it. F1 tells us that X lies somewhere in the orbit of $Y(A)$. Therefore, under unrestricted Γ, the orbit is the finest object that can be recovered uniformly. F4 formalizes which targets depend only on this orbit.

F4. Invariant target

$$\tau : \mathcal{Y} \to \mathcal{S}, \qquad \tau(g \cdot y) = \tau(y) \quad \text{for all } g \in G,\ y \in \mathcal{Y}.$$

Read as: A target τ is invariant when moving or rotating the input cannot change its value.

Simple example. Let $\tau(y)$ be the number of bright pixels. Translation and quarter-turn rotation preserve it. The horizontal coordinate of the centroid is not invariant.

Relationship and why the paper needs it. F3 defines an orbit; F4 says τ is constant on every orbit. Theorem 2.1 uses this condition to draw the exact boundary between uniformly observable and nonobservable targets.

F5. Observed data

$$O = (C, A, X).$$

Read as: One observed unit contributes pretreatment covariates C, treatment A, and transformed outcome X.

Simple example. A row may contain batch $C = 3$, reagent $A = 1$, and the acquired six-channel image X.

Relationship and why the paper needs it. The potential outcomes and Γ are latent; only O is seen. Section 3 will distinguish this full observed datum from its quotient-reduced version.

Why observing X does not mean that the causal problem has been solved

Observing X does not mean that the causal problem has been solved. There are two separate missing pieces:

1. X is a transformed measurement, not necessarily the intrinsic outcome.
2. We observe only the outcome under the treatment actually received, not the counterfactual outcome under the other treatment.

1. What exactly is observed? The observed record is

$$O = (C, A, X).$$

Read as: One observed unit contributes the covariates C, its realized treatment A, and its recorded transformed outcome X.

But the observation model is

$$X = \Gamma \cdot Y(A).$$

Read as: The recorded outcome is obtained by applying the unknown transformation Γ to the intrinsic potential outcome corresponding to the treatment actually received.

Therefore, more explicitly,

$$O = \big(C, A, \Gamma \cdot Y(A)\big).$$

Read as: We observe the covariates and treatment, but the outcome is the intrinsic treatment outcome after an unknown transformation Γ.

For a treated unit, $A = 1$, we observe

$$X = \Gamma \cdot Y(1).$$

Read as: For this treated unit, the recorded image is a transformed version of its intrinsic treated potential outcome.

We do not directly observe:

$$Y(1), \qquad Y(0), \qquad \Gamma.$$

Read as: These are, respectively, the treated intrinsic outcome, the untreated counterfactual outcome, and the unknown acquisition transformation; none is directly observed for this treated unit.

The first is hidden by the unknown transformation, the second is counterfactual, and the third is an unobserved acquisition mechanism.

2. Why is X not automatically the causal outcome? Suppose treatment has no biological effect:

$$Y(1) = Y(0) = y.$$

Read as: The same intrinsic object y would occur whether the unit received treatment or control.

However, suppose acquisition depends on treatment:

$$\Gamma(a) = \begin{cases} R_{0^\circ}, & a = 0, \\ R_{90^\circ}, & a = 1. \end{cases}$$

Read as: The acquisition system applies no rotation under control and a 90° rotation under treatment. This is an illustrative treatment-dependent acquisition mechanism, not an assumption imposed by the general theory.

Then, defining the treatment-indexed recorded outcome by $X(a) := \Gamma(a) \cdot Y(a)$,

$$X(a) = \begin{cases} y, & a = 0, \\ R_{90^\circ} \cdot y, & a = 1. \end{cases}$$

Read as: Control units are recorded without rotation, while treated units are recorded after a 90° rotation.

The raw recorded potential outcomes differ:

$$X(1) \neq X(0),$$

Read as: The treatment-indexed recorded images are different because their acquisition transformations are different.

Even though the intrinsic outcomes are identical:

$$Y(1) = Y(0).$$

Read as: Treatment has produced no intrinsic biological change in this example.

A raw-pixel comparison could therefore conclude that treatment caused a change, when the difference came entirely from acquisition geometry.

This is the first causal problem the paper addresses:

> *How can we separate treatment-induced intrinsic changes from treatment-dependent acquisition transformations?*

The precise problem solved by the paper The paper is not trying to obtain an outcome when none was observed. We already observe X. It solves the following more difficult problem:

> *Given only a treatment A, covariates C, and an outcome X whose coordinates have been altered by an unknown and potentially treatment-dependent transformation, which causal properties of the intrinsic outcome are still observable, and under what assumptions can their treatment-specific distributions be identified and tested?*

F6. Uniform decoder condition in Theorem 2.1

$$D(g \cdot y) = \tau(y) \qquad \text{for every } g \in G,\ y \in \mathcal{Y}.$$

Read as: A single decoder D must recover the target value of the original object from every possible transformed version of that object.

Simple example. From any translated or quarter-turned L-shape, D can count its three bright pixels. No decoder can recover the original absolute x-coordinate without knowing the translation.

Relationship and why the paper needs it. This is stronger than recovering the target on average under a chosen distribution of Γ. It must work for all g. The theorem states that F6 holds exactly when F4 holds.

F7. The constructive half of sharp observability

$$D = \tau.$$

Read as: If the target itself is invariant, use the target function as the decoder.

Simple example. To recover pixel count, simply count pixels in the observed transformed image; no registration is needed.

Relationship and why the paper needs it. F4 implies $\tau(g \cdot y) = \tau(y)$, so choosing $D = \tau$ immediately satisfies F6. This proves the easy direction of Theorem 2.1.

F8. The converse half of sharp observability

$$D(z) = \tau(z), \qquad \tau(g \cdot y) = D(g \cdot y) = \tau(y).$$

Read as: The identity transformation forces the decoder to equal the target on untransformed inputs; therefore, if the same decoder works after any g, the target must be invariant.

Simple example. If D recovered the original centroid coordinate from every translated image, then translating an image would have to leave that coordinate unchanged, which is false.

Relationship and why the paper needs it. Together, F7 and F8 prove the equivalence: uniform recoverability if and only if orbit invariance. This is the paper's first information boundary.

F9. Witness of noninvariance

$$\tau(g \cdot y) \neq \tau(y).$$

Read as: There exists at least one object and one allowed transformation that changes the proposed target.

Simple example. Let τ be the leftmost pixel coordinate. Translating one pixel right changes τ by one.

Relationship and why the paper needs it. This is the negation of F4. Corollary 2.2 uses it to construct two causal data-generating processes with exactly the same observation but different target values.

F10. The two indistinguishable processes in Corollary 2.2

$$\begin{array}{lll} \text{Process I:} & Y(a) = y, & \Gamma = g, \\ \text{Process II:} & Y(a) = g \cdot y, & \Gamma = e, \end{array} \qquad X = g \cdot y \text{ in both processes.}$$

Read as: In the first world, the intrinsic object is y and acquisition moves it; in the second, the intrinsic object is already moved and acquisition does nothing. Both produce the same recorded object.

Simple example. A cell centered at 0 shifted to 5 and a cell intrinsically centered at 5 with no shift both yield an image centered at 5.

Relationship and why the paper needs it. F9 guarantees that the two worlds have different $\tau\{Y(a)\}$ values. Since their observed laws agree, no observed-data method can identify a noninvariant target under F1 without extra assumptions.

3.3 Section 2.2: maximal invariance and factorization

F11. Maximal invariant (Definition 2.3)

$$M(y) = M(z) \iff z = g \cdot y \text{ for some } g \in G.$$

Read as: Two objects receive the same summary M exactly when they differ only by an allowed transformation.

Simple example. Canonicalize every L-shape by moving its bounding box to the origin and selecting the lexicographically smallest quarter-turn. All moved copies have one canonical bitmap; a square has another.

Relationship and why the paper needs it. Ordinary invariance supplies only the forward implication. Maximality adds the reverse implication, so M neither distinguishes equivalent objects nor merges genuinely different orbits. It is the complete observable quotient.

F12. Borel factorization (Theorem 2.4)

$$h = \bar{h} \circ M.$$

Read as: Every measurable invariant target h can be computed in two stages: first compute the maximal invariant M, then apply a measurable function $\bar{h}$ to it.

Simple example. If $M(y)$ is the canonical image and $h(y)$ is bright-pixel count, then $\bar{h}$ simply counts pixels in the canonical image.

Relationship and why the paper needs it. F11 says M labels complete orbits; F12 says it retains every measurable invariant target. This justifies using M as a universal observable representation, but it does not yet say that $M(X)$ is statistically sufficient for parameters governing the raw law.

3.4 Section 2.3: causal identification of the quotient law

F13. Quotient potential outcome

$$Q(a) = M\{Y(a)\}.$$

Read as: Under treatment a, keep the intrinsic structured outcome only up to the allowed transformations.

Simple example. $Q(0)$ is the canonical L-shape and $Q(1)$ is the canonical square, regardless of where or how either would be recorded.

Relationship and why the paper needs it. F11 provides the representation M; F13 applies it to each potential outcome. This becomes the causal estimand that remains observable under unrestricted contamination.

F14. Observed quotient outcome

$$Q = M(X).$$

Read as: Compute the quotient-valued observed outcome by applying the maximal invariant to the recorded image.

Simple example. Canonicalizing the shifted, rotated acquired image returns its orbit label.

Relationship and why the paper needs it. F13 is potential and treatment-specific; F14 is observed. Formula F15 connects them through consistency.

F15. Quotient consistency (paper Eq. 2.1)

$$Q = M(X) = M\{\Gamma \cdot Y(A)\} = M\{Y(A)\} = Q(A).$$

Read as: Canonicalizing the observation removes the unknown transformation, leaving the quotient potential outcome corresponding to the treatment actually received.

Simple example. If treatment 1 produces a square and acquisition rotates it, both the rotated square and intrinsic square canonicalize to the same $Q(1)$.

Relationship and why the paper needs it. The first equality is F14, the second inserts F1, the third uses invariance of M, and the last uses F13. It is the bridge from the observation model to causal potential-outcome notation, and it requires no distributional assumption on Γ.

F16. Conditional exchangeability

$$Q(a) \perp\!\!\!\perp A \mid C.$$

Read as: Within levels of pretreatment covariates C, actual treatment assignment carries no additional information about the quotient potential outcome under treatment a.

Simple example. Within each randomized plate, which well receives reagent 1 is independent of the wells' canonical potential morphologies.

Relationship and why the paper needs it. F15 alone connects observed and potential quotients only for the received treatment. F16 allows information from units with $A = a$ to represent the counterfactual quotient outcomes of comparable units.

F17. Positivity

$$\mathbb{P}(A = a \mid C) > 0 \qquad \text{almost surely.}$$

Read as: Every covariate stratum that occurs must have a positive chance of receiving treatment a.

Simple example. If both reagents can occur on every relevant plate, positivity holds. If one plate can receive only control, its treated law cannot be learned from that plate.

Relationship and why the paper needs it. Exchangeability F16 is useless in a stratum with no treated observations. Positivity ensures the conditional law in F18 exists on the strata that must be averaged.

F18. Adjustment for the full quotient law (paper Eq. 2.2)

$$\mathbb{P}\{Q(a) \in B\} = \int \mathbb{P}(Q \in B \mid A = a, C = c)\, dP_C(c), \qquad B \subseteq \mathcal{S} \text{ Borel.}$$

Read as: To find the probability that the treatment-a quotient outcome falls in set B, calculate that probability among observed units receiving a within each covariate level, then average over the population distribution of covariates.

Simple example. Suppose half the population is batch 1 and half batch 2; treated canonical squares occur with probabilities 0.8 and 0.6. The adjusted probability is $0.5(0.8) + 0.5(0.6) = 0.7$.

Relationship and why the paper needs it. F15 supplies consistency, F16 permits replacement of potential by observed conditional laws, and F17 makes those laws observable. Requiring F18 for every Borel B identifies the entire distribution, not merely a mean.

F19. Pushforward of any invariant causal target

$$h = \bar{h} \circ M \quad \Longrightarrow \quad \mathrm{Law}\{h(Y(a))\} = \bar{h}_{\#}\, \mathrm{Law}\{Q(a)\}.$$

Read as: Once the law of the maximal invariant is known, the law of any invariant target is obtained by applying its factor map $\bar{h}$.

1. What are the objects?

The maps and random variables have the following order:

$$Y(a) \xrightarrow{M} Q(a) = M\{Y(a)\} \xrightarrow{\bar{h}} \bar{h}\{Q(a)\} = h\{Y(a)\}.$$

Read as: Start with the intrinsic image under treatment a, replace it by its quotient or canonical representative, and then apply $\bar{h}$ to obtain the invariant target value.

Here:

- $Y(a)$ is the intrinsic image under treatment a;
- M removes the distinctions caused only by irrelevant translation and rotation by mapping every orbit to one quotient label;
- $Q(a) = M\{Y(a)\}$ is the quotient outcome—represented by the canonical image in the lattice-image implementation;
- h is an invariant target calculated from the original image, such as pixel-based cell area; and
- $\bar{h}$ is the corresponding target function calculated from the quotient or canonical image.

Thus,

$$h = \bar{h} \circ M \qquad \text{means} \qquad h(y) = \bar{h}\{M(y)\}.$$

Read as: To calculate the invariant target from y, first canonicalize y using M, and then calculate the target using $\bar{h}$.

For example, when each foreground pixel has unit area,

$$\text{pixel-based cell area of the original image} = \text{pixel count of its canonical image.}$$

Read as: Translation and rotation change the coordinates of the foreground pixels but not how many foreground pixels there are.

2. What does $\bar{h}_\#$ mean?

Let
$$P_Q = \mathrm{Law}\{Q(a)\}.$$

Read as: P_Q is the treatment-a probability distribution of the quotient or canonical image.

Then
$$\bar{h}_\# P_Q$$

Read as: This is pronounced "the pushforward of P_Q by $\bar{h}$": move the probability distribution on quotient images through the function $\bar{h}$ to obtain a distribution on target values.

For any possible measurable set B of target values,
$$(\bar{h}_\# P_Q)(B) = P_Q\{\bar{h}^{-1}(B)\}.$$

Read as: The probability assigned to target-value set B equals the probability of all quotient images whose $\bar{h}$ values fall inside B.

Equivalently,
$$(\bar{h}_\# P_Q)(B) = \mathbb{P}\{\bar{h}(Q(a)) \in B\}.$$

Read as: To find the probability that the transformed value belongs to B, collect every canonical image that $\bar{h}$ sends into B and add their probabilities.

The inverse image is
$$\bar{h}^{-1}(B) = \{q : \bar{h}(q) \in B\}.$$

Read as: The inverse image means "all inputs whose outputs belong to B"; it does not require $\bar{h}$ to have an inverse function.

3. Detailed numerical example

Suppose that, under treatment $a = 1$, the canonical image $Q(1)$ can take three possible forms:

Canonical image	Probability	Pixel count
q_1	0.50	10
q_2	0.30	10
q_3	0.20	20

Therefore,
$$\mathbb{P}\{Q(1) = q_1\} = 0.50, \qquad \mathbb{P}\{Q(1) = q_2\} = 0.30, \qquad \mathbb{P}\{Q(1) = q_3\} = 0.20.$$

Read as: Under treatment 1, canonical images q_1, q_2, and q_3 occur with probabilities 0.50, 0.30, and 0.20, respectively.

Define $\bar{h}$ as pixel count:
$$\bar{h}(q_1) = 10, \qquad \bar{h}(q_2) = 10, \qquad \bar{h}(q_3) = 20.$$

Read as: Although q_1 and q_2 are different canonical shapes, both have pixel-based area 10, while q_3 has area 20.

Probability that the area equals 10. Take

$$B = \{10\}.$$

Read as: The set B contains the one target value of interest: area 10.

The inverse image is

$$\bar{h}^{-1}(\{10\}) = \{q_1, q_2\}.$$

Read as: The canonical images mapped to area 10 are q_1 and q_2.

Therefore,

$$\begin{aligned}(\bar{h}_{\#}P_Q)(\{10\}) &= P_Q\{\bar{h}^{-1}(\{10\})\} \\ &= P_Q(\{q_1, q_2\}) \\ &= 0.50 + 0.30 = 0.80.\end{aligned}$$

Read as: After pushing the canonical-image distribution through pixel count, the probability of cell area 10 is 0.80.

Similarly,

$$\bar{h}^{-1}(\{20\}) = \{q_3\}, \qquad (\bar{h}_{\#}P_Q)(\{20\}) = 0.20.$$

Read as: Only q_3 maps to area 20, so the pushed-forward distribution assigns area 20 probability 0.20.

The resulting area distribution is therefore:

Cell area	Probability
10	0.80
20	0.20

This is precisely

$$\mathrm{Law}\{h(Y(1))\}.$$

Read as: The pushed-forward probabilities are exactly the treatment-1 distribution of the invariant area measured on the intrinsic image.

4. Why are the two sides equal?

Since

$$Q(a) = M\{Y(a)\},$$

Read as: The quotient potential outcome is the maximal-invariant representation of the intrinsic potential outcome.

and

$$h = \bar{h} \circ M,$$

Read as: The invariant target h factors through the maximal invariant M using $\bar{h}$.

we obtain

$$\begin{aligned}h\{Y(a)\} &= (\bar{h} \circ M)\{Y(a)\} \\ &= \bar{h}\big(M\{Y(a)\}\big) \\ &= \bar{h}\{Q(a)\}.\end{aligned}$$

Read as: Computing h directly on the intrinsic outcome gives the same random value as first quotienting the outcome and then applying $\bar{h}$.

Therefore,

$$\mathrm{Law}\{h(Y(a))\} = \mathrm{Law}\{\bar{h}(Q(a))\}.$$

Read as: Equal random values have equal probability distributions.

By the definition of pushforward,

$$\mathrm{Law}\{\bar{h}(Q(a))\} = \bar{h}_{\#}\, \mathrm{Law}\{Q(a)\}.$$

Read as: The distribution of a function of a random variable is the pushforward of that random variable's distribution through the function.

Hence,

$$\mathrm{Law}\{h(Y(a))\} = \bar{h}_{\#}\, \mathrm{Law}\{Q(a)\}.$$

Read as: The treatment-specific law of the invariant target is obtained by pushing the identified treatment-specific quotient law through $\bar{h}$.

5. Why does the paper need this formula?

The complete intrinsic-image law

$$\mathrm{Law}\{Y(a)\}$$

Read as: This is the full treatment-specific distribution of intrinsic images, including their original positions and orientations.

is generally not identifiable because unknown translations and rotations contaminate the image coordinates. However, under the causal assumptions in F15–F18, the paper identifies the quotient law

$$\mathrm{Law}\{Q(a)\}.$$

Read as: This is the identifiable treatment-specific distribution of intrinsic images after differences due only to the allowed transformations have been removed.

F19 shows that this is enough to identify every measurable invariant target derived from the images. For example:

$$\mathrm{Law}\{\text{cell area under treatment } a\},$$
$$\mathrm{Law}\{\text{number of components under treatment } a\},$$
$$\mathrm{Law}\{\text{rotation-invariant morphology score under treatment } a\}.$$

Read as: Once the quotient distribution is known, the entire treatment-specific distribution of any measurable translation-and-rotation-invariant summary can be derived from it.

Relationship and why the paper needs it. This combines the factorization in F12 with identification of the full quotient law in F18. It explains why identifying that full law is stronger than identifying only one preselected invariant summary.

The corrected intuitive reading is:

> Once the treatment-specific distribution of canonical images is identified, apply $\bar{h}$ to every canonical image. The resulting distribution is the treatment-specific distribution of the invariant target h. The symbol # denotes this transfer of probability from canonical images to target values.

3.5 Section 3: when quotient reduction is statistically lossless

F20. Full and quotient observations

$$O_{\text{full}} = (C, A, X), \qquad O_{\text{quot}} = T(O_{\text{full}}) = (C, A, M(X)).$$

Read as: The full experiment sees raw transformed data; the quotient experiment keeps the same covariates and treatment but replaces X by its orbit label.

Simple example. One dataset stores the moved image; the reduced dataset stores its canonical bitmap together with the same batch and reagent.

Relationship and why the paper needs it. Earlier formulas asked what intrinsic targets are observable. F20 starts a stronger comparison: whether the reduced data preserve all statistical information in the raw observed-data family.

F21. Families of statistical experiments

$$\mathcal{E}_{\text{full}} = \{P_\theta^{\text{full}} : \theta \in \Theta\}, \qquad \mathcal{E}_{\text{quot}} = \{P_\theta^{\text{quot}} : \theta \in \Theta\}.$$

Read as: For every possible parameter θ, consider the law of the full data and the corresponding law of the quotient data.

Simple example. θ might govern both intrinsic morphology and how acquisition rotations are distributed. Raw orientation frequencies may then reveal θ even when all quotient images are identical.

Relationship and why the paper needs it. The distinction matters because maximal invariance concerns intrinsic orbit information, whereas statistical sufficiency concerns every feature of the parameterized raw observed law, including within-orbit contamination behavior.

F22. Quotient-faithful reconstruction kernel

$$K\big(s, T^{-1}(\{s\})\big) = 1 \quad P_\theta^{\text{quot}}\text{-almost surely for every } \theta \in \Theta.$$

Read as: Given quotient datum s, the reconstruction procedure K must generate a full datum that maps back to exactly that same s with probability one.

Simple example. If s contains a canonical L-shape, K may randomly rotate it, but it may not output a square or change its batch or treatment.

Relationship and why the paper needs it. A reverse simulator that merely reproduces the correct marginal raw distribution could scramble quotient labels. F22 rules that out and is essential to equate reconstruction with conditioning on the supplied quotient.

F23. Common conditional law

$$K(s, \cdot) = \mathrm{Law}_\theta(O_{\text{full}} \mid O_{\text{quot}} = s) \quad P_\theta^{\text{quot}}\text{-almost surely.}$$

Read as: The same kernel K, independent of θ, must describe how raw data are distributed inside each quotient fibre.

Simple example. Given a canonical image, suppose raw orientation is uniformly one of four rotations for every θ. Then one common K works. If the orientation probabilities depend on θ, it does not.

Relationship and why the paper needs it. F22 constrains where K can output; F23 says it is the actual conditional distribution and is parameter-free. Theorem 3.2 shows this is exactly the boundary for lossless quotienting.

F24. Reconstruction identity (paper Eq. 3.1)

$$P_\theta^{\mathrm{quot}} K = P_\theta^{\mathrm{full}} \qquad \text{for every } \theta \in \Theta.$$

Read as: Sample a quotient datum from its law, then reconstruct with K; the resulting full datum has exactly the original raw-data law for every parameter value.

Simple example. Draw a canonical L-shape, then apply a rotation according to the common within-orbit distribution. If this reproduces the observed raw-image distribution for every θ, F24 holds.

Relationship and why the paper needs it. F23 implies F24 by integrating conditional distributions. Conversely, F24 plus quotient faithfulness F22 implies F23. This equivalence is the mathematical core of Theorem 3.2.

F25. Sufficiency and Blackwell equivalence

$$O_{\mathrm{quot}} \text{ is sufficient for } \mathcal{E}_{\mathrm{full}}, \qquad \mathcal{E}_{\mathrm{full}} \equiv_B \mathcal{E}_{\mathrm{quot}}.$$

Read as: When F22–F24 hold, the quotient and raw experiments support exactly the same statistical decisions and attainable risks.

Simple example. No estimator or classifier with a bounded loss can improve by seeing raw orientation once the canonical image is known, because the raw orientation can be regenerated without knowing θ.

Relationship and why the paper needs it. The deterministic map T always makes the full data at least as informative as the quotient. The parameter-free K supplies the reverse direction. This claim is much stronger than F12.

F26. Disintegration identity used in the first direction

$$Q_\theta := P_\theta^{\mathrm{quot}}, \qquad P_\theta^{\mathrm{full}}(B) = \int K(s,B)\, Q_\theta(ds) = (Q_\theta K)(B).$$

Read as: The unconditional probability of a raw event B is obtained by averaging its conditional probability given the quotient over the quotient distribution.

Simple example. If 60% of quotients are L-shapes and K assigns a 90° raw orientation with probability $1/4$, then the joint probability of L-shape and that orientation is $0.6/4 = 0.15$.

Relationship and why the paper needs it. This is the law of total probability for a Markov kernel. It proves that the common conditional law F23 reconstructs the full law F24.

F27. Fibre-intersection identity used in the converse

$$K\big(s, B \cap T^{-1}(D)\big) = \mathbf{1}_D(s) K(s,B).$$

Read as: If reconstruction preserves the supplied quotient, requiring the reconstructed datum also to have quotient in D either changes nothing when $s \in D$, or makes the event impossible when $s \notin D$.

Simple example. If D is the set of square quotients and s is an L-shape quotient, the left side is zero. If s is a square, it equals $K(s,B)$.

Relationship and why the paper needs it. F27 is the operational consequence of quotient faithfulness F22. It lets F24 be converted into the defining conditional-probability identity in F28.

F28. Joint-event reconstruction

$$\begin{aligned}P_\theta^{\text{full}}\big(B \cap T^{-1}(D)\big) &= \int K\big(s, B \cap T^{-1}(D)\big) Q_\theta(ds) \\ &= \int_D K(s, B) Q_\theta(ds).\end{aligned}$$

Read as: The probability that a raw event B occurs together with a quotient event D equals the average conditional probability of B over quotient values inside D.

Simple example. Sum the probabilities of a selected raw-orientation event only across canonical shapes belonging to D.

Relationship and why the paper needs it. The first line uses F24; the second uses F27. It establishes that K behaves as a regular conditional distribution, not merely as a marginal-law simulator.

F29. Defining identity for the conditional law

$$P_\theta^{\text{full}}(O_{\text{full}} \in B, O_{\text{quot}} \in D) = \int_D K(s, B) P_\theta^{\text{quot}}(ds).$$

Read as: To obtain a joint probability, integrate the conditional chance of the full-data event over the chosen quotient values.

Simple example. For $D =$ square shapes and $B =$ raw images facing up, the formula averages the facing-up probability among square quotients.

Relationship and why the paper needs it. F29 is exactly the formal definition needed to recover F23 from F22 and F24. This completes the equivalence in Theorem 3.2.

F30. The two Blackwell comparisons

$$\mathcal{E}_{\text{full}} \succeq_B \mathcal{E}_{\text{quot}}, \qquad \mathcal{E}_{\text{quot}} \succeq_B \mathcal{E}_{\text{full}}.$$

Read as: The first experiment can simulate the second through deterministic quotienting, and the second can simulate the first through K.

Simple example. Raw data can always be canonicalized. Under a common within-orbit law, canonical data can also be randomized back into raw data.

Relationship and why the paper needs it. Having both comparisons yields the equivalence in F25. Without the second, quotienting can be strictly less informative.

F31. Counterexample: maximal invariance need not imply sufficiency

$$G = \{-1, 1\}, \quad Y = 1, \quad \mathbb{P}_\theta(\Gamma = 1) = \theta, \quad X = \Gamma, \quad \mathbb{P}_\theta(X = 1) = \theta.$$

Read as: The intrinsic outcome never changes, but the probability of observing raw sign $+1$ equals the parameter θ because the contamination sign carries it.

Simple example. With $\theta = 0.9$, 90% of raw observations are $+1$; with $\theta = 0.2$, only 20% are. The orbit under multiplication by ± 1 is the same in both cases.

Relationship and why the paper needs it. The maximal invariant is constant and hence has the same law for every θ, while raw X identifies θ. This proves why F12 cannot replace the stronger common-kernel condition F23.

3.6 Section 3.1: conditional Haar contamination

F32. Attained quotient space

$$\mathcal{S}_0 = M(\mathcal{Y}).$$

Read as: $\mathcal{S}_0$ is the set of quotient labels that actually come from objects in $\mathcal{Y}$.

Simple example. If M outputs canonical finite images, $\mathcal{S}_0$ contains the canonical images that can really occur, not every formal array in a larger ambient space.

Relationship and why the paper needs it. Reconstruction only needs to be defined meaningfully on attained quotients. This avoids imposing behavior on irrelevant $s \notin \mathcal{S}_0$.

F33. Measurable section

$$s : \mathcal{S}_0 \to \mathcal{Y}, \qquad M\{s(m)\} = m \quad \text{for every } m \in \mathcal{S}_0.$$

Read as: For each orbit label m, choose one measurable representative $s(m)$ that belongs to that orbit.

Simple example. For an orbit of moved L-shapes, choose the L-shape whose bounding box starts at the origin and whose serialized pixels are lexicographically smallest.

Relationship and why the paper needs it. A quotient label is not itself necessarily a raw outcome. F33 supplies a representative to which random group transformations can be applied in the orbit kernel F34.

F34. Haar orbit kernel (paper Eq. 3.2)

$$K_m(B) = \int_G \mathbf{1}_B\{g \cdot s(m)\}\, d\mu_G(g).$$

Read as: To generate a raw representative from orbit m, draw g from the uniform Haar distribution on the compact group and ask whether $g \cdot s(m)$ falls in event B.

Simple example. For the four rotations C_4, $K_m(B)$ is the fraction of the four rotated versions of $s(m)$ that lie in B.

Relationship and why the paper needs it. F33 provides the starting representative; Haar invariance ensures the resulting distribution does not depend on which representative was selected. F34 is the concrete candidate for the common kernel required by F23.

F35. The quotient is attained almost surely

$$M(X) \in \mathcal{S}_0 \qquad \text{almost surely.}$$

Read as: Every quotient computed from an actual observation belongs to the range of M.

Simple example. Canonicalizing a real image cannot produce a quotient label that no image can attain.

Relationship and why the paper needs it. This is why the extension of K_m outside $\mathcal{S}_0$ can be arbitrary: those inputs are never encountered under the experiment.

F36. Conditional Haar assumption (paper Eq. 3.3)

$$\mathrm{Law}_\theta\{\Gamma \mid C, A, Y(A)\} = \mu_G \qquad \text{almost surely for every } \theta.$$

Read as: Even after knowing covariates, treatment, and the intrinsic realized outcome, the nuisance transformation is uniformly distributed on the compact group, with a law that does not depend on θ.

Simple example. Conditional on every image and treatment, each of four rotations occurs with probability $1/4$.

Relationship and why the paper needs it. The paper does *not* assume F36 in its main unrestricted model. It introduces F36 only as a sufficient special case that makes the within-orbit conditional law parameter-free.

F37. Parameter-free conditional raw law under Haar contamination

$$\mathrm{Law}_\theta\{X \mid C, A, M(X) = m\} = K_m.$$

Read as: Once covariates, treatment, and the orbit label are known, the remaining raw variation has the Haar orbit distribution and no longer depends on θ.

Simple example. Given the canonical L-shape, the raw image is uniformly one of its four orientations, whatever the biological parameter is.

Relationship and why the paper needs it. F36 implies F37. Formula F37 is precisely the common conditional-law requirement F23, specialized to the image coordinate.

F38. Blackwell consequence of the Haar condition

$$\mathcal{E}_{\text{full}} \equiv_B \mathcal{E}_{\text{quot}}.$$

Read as: Under F36 and the measurability assumptions, raw and quotient data are equally informative for all bounded decision problems.

Simple example. The raw rotation can be resimulated uniformly from the canonical image, so retaining it cannot help estimate θ.

Relationship and why the paper needs it. F37 plugs directly into Theorem 3.2, giving F38. This is a corollary, not the foundation of observability or quotient identification.

F39. Conditional probability before quotient conditioning

$$\mathbb{P}_\theta\{X \in B \mid C, A, Y(A)\} = \int_G \mathbf{1}_B\big(g \cdot Y(A)\big)\, d\mu_G(g).$$

Read as: Given the intrinsic outcome, average the event indicator over all group transformations according to Haar probability.

Simple example. For four rotations, add four zeros or ones indicating whether each rotated image belongs to B, then divide by four.

Relationship and why the paper needs it. This is F1 combined with the conditional law F36. The next formulas show that this average depends only on the orbit $M\{Y(A)\}$.

F40. Relating an arbitrary representative to the section

$$m = M(y), \qquad y = h \cdot s(m) \quad \text{for some } h \in G.$$

Read as: If m is the orbit label of y, then y can be obtained from the selected representative $s(m)$ by some group transformation h.

Simple example. A translated, rotated L-shape y is some motion h applied to its canonical representative.

Relationship and why the paper needs it. This uses maximality F11 and section property F33. It lets the integral around y be rewritten around $s(m)$.

F41. Change of representative inside the Haar integral

$$\int_G \mathbf{1}_B(g \cdot y)\, d\mu_G(g) = \int_G \mathbf{1}_B\big((gh) \cdot s(m)\big)\, d\mu_G(g).$$

Read as: Replace y by $h \cdot s(m)$; applying g to it is the same as applying the composed transformation gh to $s(m)$.

Simple example. Rotating an already 90°-rotated image by 180° is the same as applying the composed 270° rotation to the canonical image.

Relationship and why the paper needs it. F41 is group associativity applied to F40. Right invariance of Haar measure then removes the fixed h in F42.

F42. Representative independence of the orbit kernel

$$\int_G \mathbf{1}_B(g \cdot y)\, d\mu_G(g) = \int_G \mathbf{1}_B\big(g \cdot s(m)\big)\, d\mu_G(g) = K_m(B).$$

Read as: Haar averaging around any member of an orbit gives the same probability distribution as Haar averaging around the selected representative.

Simple example. Reordering the four rotations by first adding a fixed quarter turn does not change their uniform average.

Relationship and why the paper needs it. This is the key use of right invariance of μ_G. It proves that F34 is genuinely a function of the orbit label rather than the arbitrary section.

F43. Conditional law depends only on the quotient

$$\mathbb{P}_\theta\{X \in B \mid C, A, Y(A)\} = K_{M\{Y(A)\}}(B) \qquad \text{almost surely.}$$

Read as: Given the intrinsic outcome, the event probability can be written using only its orbit label.

Simple example. Any two intrinsic images that are moved copies of the same L-shape produce the same uniform distribution over raw orientations.

Relationship and why the paper needs it. F39 gives the integral and F42 identifies it as the orbit kernel. Because the right side is measurable with respect to the quotient, conditioning down yields F44.

F44. Conditioning down to the quotient

$$\begin{aligned}\mathbb{P}_\theta\{X \in B \mid C, A, M(Y(A))\} &= \mathbb{E}_\theta\big[K_{M\{Y(A)\}}(B) \mid C, A, M(Y(A))\big] \\ &= K_{M\{Y(A)\}}(B).\end{aligned}$$

Read as: Once a quantity is already a function of the information being conditioned on, its conditional expectation equals itself.

Simple example. If the quotient says "L-shape," the conditional raw-event probability is the fixed L-shape orbit-kernel value; the unknown original representative adds nothing.

Relationship and why the paper needs it. This turns F43 into a conditional law given the quotient alone, preparing the exact common conditional law required by Theorem 3.2.

F45. Invariance identifies the latent and observed quotients

$$M(X) = M\{\Gamma \cdot Y(A)\} = M\{Y(A)\} \qquad \text{almost surely.}$$

Read as: The quotient computed from the observed image equals the quotient of the intrinsic outcome.

Simple example. Canonicalizing a rotated L-shape gives the same result as canonicalizing the unrotated L-shape.

Relationship and why the paper needs it. This repeats the essential middle part of F15 inside the Haar proof. Substituting observed $M(X)$ for latent $M(Y(A))$ converts F44 into F37.

F46. Full-data reconstruction kernel

$$\widetilde{K}\big((c, a, m), B\big) := \int_{\mathcal{Y}} \mathbf{1}_B(c, a, x) K_m(dx).$$

Read as: Keep covariates c and treatment a fixed, draw a raw outcome x from the orbit kernel K_m, and check whether the reconstructed triple lies in event B.

Simple example. Input $(\text{batch } 3, \text{treated}, \text{L-shape})$ returns the same batch and treatment plus a uniformly rotated L-shape.

Relationship and why the paper needs it. F34 reconstructs only X. F46 lifts it to a kernel from the entire quotient datum to the entire full datum, which is what Theorem 3.2 formally requires.

F47. Quotient faithfulness of the Haar reconstruction

$$M\{g \cdot s(m)\} = M\{s(m)\} = m \qquad \text{for every } g \in G.$$

Read as: Every reconstructed group transform remains in the orbit indexed by the supplied label m.

Simple example. Every rotated canonical L-shape canonicalizes back to that same L-shape.

Relationship and why the paper needs it. The first equality uses invariance; the second uses the section F33. It verifies the fibre condition F22 for $\widetilde{K}$ and completes the proof of F38.

3.7 Section 4: product versus diagonal actions

F48. Componentwise maximal-invariant map

$$M^{\times J}(y_1, \ldots, y_J) = \big(M(y_1), \ldots, M(y_J)\big), \qquad J \geq 2.$$

Read as: Canonicalize each labeled component separately and keep the ordered list of canonical results.

Simple example. For a two-site well, canonicalize site 1 and site 2 independently.

Relationship and why the paper needs it. Whether F48 is a complete invariant depends on the scientific nuisance action. F49 and F50 distinguish two fundamentally different cases.

F49. Product action

$$(g_1, \ldots, g_J) \cdot (y_1, \ldots, y_J) = \big(g_1 \cdot y_1, \ldots, g_J \cdot y_J\big).$$

Read as: Each component may undergo its own transformation.

Simple example. Site 1 can rotate 90° while site 2 rotates 270° and shifts differently.

Relationship and why the paper needs it. Under F49, relative pose is not uniformly observable because independent transformations can alter it arbitrarily. Proposition 4.1 proves F48 is maximal for this product action.

F50. Diagonal action

$$g \cdot (y_1, \ldots, y_J) = \big(g \cdot y_1, \ldots, g \cdot y_J\big).$$

Read as: One common transformation acts on every component.

Simple example. A single microscope-stage shift moves both sites by exactly the same amount.

Relationship and why the paper needs it. Under F50, relative displacement or orientation between components survives. Componentwise canonicalization F48 is invariant but may merge distinct diagonal orbits and hence fail to be maximal.

F51. Criterion for componentwise maximality under a diagonal action

$$y, z \in \mathcal{D}, \quad z_j = g_j \cdot y_j \text{ for each } j \quad \Longrightarrow \quad \exists g \in G \text{ such that } z_j = g \cdot y_j \text{ for every } j.$$

Read as: Whenever components can be matched separately, they must also be matchable by one common transformation; only then is separate canonicalization complete for a shared-action model.

Simple example. If site 1 needs a shift of $+2$ but site 2 needs -1, separate matching is possible but common matching is not. F51 fails.

Relationship and why the paper needs it. Equality of F48 outputs guarantees component-specific g_j by maximality of M. F51 states exactly when that equality also guarantees one diagonal-orbit relation.

F52. Equality of componentwise quotient labels

$$M^{\times J}(y_1,\ldots,y_J) = M^{\times J}(z_1,\ldots,z_J).$$

Read as: Each y_j and z_j has the same single-component orbit label.

Simple example. Both first sites are L-shapes up to motion, and both second sites are squares up to motion.

Relationship and why the paper needs it. Under the product action, F52 is equivalent to the same product orbit. Under a diagonal action it may be too coarse, which motivates the criterion F51.

F53. Product-orbit reconstruction in the proof

$$(z_1,\ldots,z_J) = (g_1,\ldots,g_J)\cdot(y_1,\ldots,y_J).$$

Read as: The component-specific transformations assemble into one element of the product group G^J.

Simple example. The pair consisting of a 90° rotation for site 1 and a 270° rotation for site 2 is one element of $G\times G$.

Relationship and why the paper needs it. F52 plus single-component maximality supplies each g_j, and F53 proves maximality of F48 for F49.

F54. Componentwise invariance under a common action

$$M(g\cdot y_j) = M(y_j) \qquad \text{for every } j.$$

Read as: Applying the same transformation to all components still leaves each component's invariant unchanged.

Simple example. Moving both sites right leaves both canonical images unchanged.

Relationship and why the paper needs it. This proves F48 is at least invariant under F50. It does not prove maximality because relative configuration may have been erased.

F55. Relative-displacement invariant

$$y_2 - y_1.$$

Read as: For two points undergoing a common translation, subtracting their positions cancels that translation.

Simple example. $(y_1,y_2)=(2,5)$ has difference 3. After adding 10 to both, $(12,15)$ still has difference 3.

Relationship and why the paper needs it. A one-point quotient for the transitive translation action is constant, so F48 would discard this difference. F55 is the simplest witness that product and diagonal quotients are not interchangeable.

3.8 Section 5: approximate contamination and stability

F56. Orbit pseudometric (paper Eq. 5.1)

$$d_G(y, z) = \inf_{g \in G} d(y, g \cdot z).$$

Read as: Measure how close y and z can become after optimally transforming z.

Simple example. If z is exactly a shifted version of y, choose the inverse shift and obtain $d_G(y, z) = 0$. If one pixel intensity differs after best alignment, the distance remains positive.

Relationship and why the paper needs it. Exact orbit equivalence becomes zero distance. This metric lets the paper quantify small violations of the exact contamination model rather than merely label them invariant or noninvariant.

F57. Lipschitz quotient representation (paper Eq. 5.2)

$$d_{\mathcal{S}}\{M(y), M(z)\} \le L_M d_G(y, z).$$

Read as: A small best-aligned change in the original objects causes at most a proportional change in their quotient representations.

Simple example. If $L_M = 2$ and the orbit distance is 0.05, the quotient distance is at most 0.10.

Relationship and why the paper needs it. F56 defines input error; F57 transfers it to the quotient space. The paper explicitly warns that its lexicographic canonicalizer may not satisfy this globally near ties or support changes.

F58. Kernel mean embedding

$$\mu_k(P) = \int_{\mathcal{S}} \Phi(s)\, P(ds).$$

Read as: Map each quotient outcome into the kernel feature space and average those feature vectors under distribution P.

Simple example. For a discrete law with two quotient images of probabilities 0.7 and 0.3, the embedding is $0.7\Phi(s_1) + 0.3\Phi(s_2)$.

Relationship and why the paper needs it. Wasserstein distance will control distributional movement in the quotient space, while F58 lets that movement be related to the kernel statistic used in the experiment.

F59. Maximum mean discrepancy

$$\mathrm{MMD}_k(P, Q) = \|\mu_k(P) - \mu_k(Q)\|_{\mathcal{H}_k}.$$

Read as: MMD is the distance between the average kernel-feature representations of two probability laws.

Simple example. If treated and control quotient distributions have identical embeddings, MMD is zero; with a characteristic kernel this implies the laws themselves are equal.

Relationship and why the paper needs it. F58 defines each mean embedding. F59 is both the population contrast bounded in Theorem 5.1 and the basis of the empirical statistic in Section 7.

F60. Approximate-contamination coupling assumption

$$\mathbb{E}\, d_G(X_a, Y_a) \leq \varepsilon_a, \qquad a \in \{0, 1\}.$$

Read as: Couple the approximately contaminated observation X_a and intrinsic outcome Y_a so that their expected orbit mismatch in arm a is at most ε_a.

Simple example. If an off-grid 5° rotation cannot be exactly undone by quarter turns but leaves average best-orbit error 0.08, take $\varepsilon_a = 0.08$.

Relationship and why the paper needs it. F60 quantifies the deviation from exact group contamination. Together with F57, it bounds the induced difference between quotient laws.

F61. Intrinsic and approximately contaminated quotient laws

$$\widetilde{P}_a = \mathrm{Law}\{M(X_a)\}, \qquad P_a = \mathrm{Law}\{M(Y_a)\}.$$

Read as: $\widetilde{P}_a$ is the quotient law actually obtained under approximate contamination, while P_a is the desired intrinsic quotient law.

Simple example. One is the distribution of canonicalized 5°-perturbed images; the other is the distribution of canonicalized clean images.

Relationship and why the paper needs it. These are the two distributions compared arm by arm in F62 and across arms in F64.

F62. Wasserstein stability (paper Eq. 5.3)

$$W_{1,d_{\mathcal{S}}}(\widetilde{P}_a, P_a) \leq L_M \varepsilon_a.$$

Read as: The quotient distribution moves by no more than the representation Lipschitz constant times the average orbit error.

Simple example. With $L_M = 2$ and $\varepsilon_a = 0.05$, the Wasserstein error is at most 0.10.

Relationship and why the paper needs it. F60 controls paired input error and F57 transfers it through M. Since any coupling upper-bounds Wasserstein distance, the result follows directly.

F63. Lipschitz kernel feature map (paper Eq. 5.4)

$$\|\Phi(s) - \Phi(t)\|_{\mathcal{H}_k} \leq L_k d_{\mathcal{S}}(s, t).$$

Read as: Nearby quotient points must have nearby feature vectors, with proportionality constant L_k.

Simple example. If $L_k = 3$ and $d_{\mathcal{S}}(s, t) = 0.02$, their feature distance is at most 0.06.

Relationship and why the paper needs it. F57 controls the quotient representation; F63 controls the next map into kernel space. Both are needed to bound the treatment-control MMD perturbation.

F64. MMD stability (paper Eq. 5.5)

$$\left|\mathrm{MMD}_k(\widetilde{P}_1, \widetilde{P}_0) - \mathrm{MMD}_k(P_1, P_0)\right| \le L_k L_M(\varepsilon_1 + \varepsilon_0).$$

Read as: The error in the observed between-arm MMD is bounded by the sum of the arm-specific contamination errors, scaled by the quotient and feature-map sensitivities.

Simple example. If $L_M = 2$, $L_k = 1$, and both $\varepsilon_a = 0.05$, the MMD changes by at most 0.20.

Relationship and why the paper needs it. F62 handles one arm at a time; F63 transfers each arm's error to its mean embedding. The reverse triangle inequality then adds the two arm errors to obtain F64.

F65. Coupling proof of Wasserstein stability

$$W_{1,d_{\mathcal{S}}}(\widetilde{P}_a, P_a) \le \mathbb{E}\, d_{\mathcal{S}}\{M(X_a), M(Y_a)\} \le L_M \varepsilon_a.$$

Read as: Use the supplied pair (X_a, Y_a) as a candidate transport coupling, then apply the Lipschitz bound F57 and expected-error bound F60.

Simple example. Transport every perturbed image's quotient to its clean partner's quotient; the average cost upper-bounds optimal transport cost.

Relationship and why the paper needs it. This is the proof mechanism behind F62 and shows exactly where the coupling assumption enters.

F66. Mean-embedding perturbation

$$\mu(R) = \mathbb{E}_{S\sim R}\Phi(S), \qquad \|\mu(\widetilde{P}_a) - \mu(P_a)\|_{\mathcal{H}_k} \le L_k L_M \varepsilon_a.$$

Read as: A distribution's mean feature is its expected feature vector, and approximate contamination changes that vector by at most the composed Lipschitz bound.

Simple example. Pair each perturbed quotient feature with its clean feature, bound each difference, then average.

Relationship and why the paper needs it. F66 combines F63 with the same coupling used in F65. Applying it for arms 0 and 1 yields F64 through the reverse triangle inequality.

F67. Gaussian kernel and its feature-distance bound

$$k(s,t) = \exp\left\{-\frac{\|s-t\|^2}{2\sigma^2}\right\}, \qquad \|\Phi(s) - \Phi(t)\|^2 = 2\{1 - k(s,t)\} \le \frac{\|s-t\|^2}{\sigma^2}.$$

Read as: Gaussian similarity decays with squared distance; the associated feature vectors are at most $1/\sigma$ times as far apart as the original points.

Simple example. If $\sigma = 2$ and $\|s - t\| = 0.2$, their feature distance is at most 0.1.

Relationship and why the paper needs it. The inequality verifies F63 with $L_k = 1/\sigma$ for a Hilbert quotient embedding. It does not verify the separate quotient-Lipschitz condition F57 for the lexicographic canonicalizer.

3.9 Section 6.1: lattice rigid motions and canonicalization

F68. Four quarter turns

$$C_4 = \{0, 1, 2, 3\}, \qquad r + s \text{ is computed modulo } 4, \qquad R_r : \mathbb{Z}^2 \to \mathbb{Z}^2.$$

Read as: The rotation index r records r counterclockwise quarter turns on the integer lattice; after four turns one returns to the starting orientation.

Simple example. $r = 1$ is 90°, $r = 3$ is 270°, and $3 + 2 \equiv 1 \pmod 4$.

Relationship and why the paper needs it. Restricting to quarter turns keeps lattice pixels on lattice pixels. This finite residual rotation group can be searched exactly after translations are normalized.

F69. Lattice rigid-motion group

$$G_{\text{rig}} = \mathbb{Z}^2 \rtimes C_4.$$

Read as: A lattice rigid motion consists of an arbitrary integer translation together with a quarter-turn rotation, combined as a semidirect product.

Simple example. $((5, -2), 1)$ means shift by $(5, -2)$ and rotate by 90° according to the action convention.

Relationship and why the paper needs it. The product is semidirect rather than direct because rotating changes how translation vectors compose. F70 makes that dependence explicit.

F70. Group multiplication and inverse

$$(t, r)(u, s) = \big(t + R_r u, r + s\big), \qquad (t, r)^{-1} = \big(-R_r^{-1} t, -r\big).$$

Read as: To compose two motions, rotate the second translation into the first coordinate frame before adding it; the inverse undoes rotation and the appropriately rotated translation.

Simple example. If the first motion rotates by 90°, a subsequent translation $u = (1, 0)$ contributes $R_1 u = (0, 1)$ in the composed translation.

Relationship and why the paper needs it. F70 guarantees that the actions used in canonicalization form a group and supplies the inverse used in the maximality proof F82.

F71. Action on an image (paper Eq. 6.1)

$$\{(t, r) \cdot x\}(v) = x\{R_r^{-1}(v - t)\}, \qquad v \in \mathbb{Z}^2.$$

Read as: To find the transformed image value at output pixel v, subtract the translation and rotate backward to locate the corresponding input pixel.

Simple example. For a pure shift $t = (2, 0)$ and $r = 0$, output location $v = (5, 3)$ receives the old value at $(3, 3)$.

Relationship and why the paper needs it. This pullback formula prevents holes and defines precisely how F69 acts on image functions. It instantiates the abstract action F2.

F72. Translation normalization

$$b(x) = \text{lower corner of the support bounding box}, \qquad N(x) = (-b(x), 0) \cdot x.$$

Read as: Locate the lower corner of the nonzero support and translate it to the origin.

Simple example. If the nonzero support begins at $(5, -2)$, apply translation $(-5, 2)$.

Relationship and why the paper needs it. The translation group $\mathbb{Z}^2$ is noncompact, so the paper does not average over it. F72 provides an explicit cross-section that removes it deterministically.

F73. Four normalized rotation candidates

$$x_r = N\{(0, r) \cdot x\}, \qquad r \in C_4.$$

Read as: Rotate x by each quarter turn and retranslate each rotated support to the origin.

Simple example. Produce the 0°, 90°, 180°, and 270° versions of the L-shape, each anchored at its bounding-box corner.

Relationship and why the paper needs it. F72 removes translation; F73 enumerates the remaining finite rotation ambiguity. F74 chooses a unique candidate.

F74. Canonical representative (paper Eq. 6.2)

$$c(x) = x_{r^\star}, \qquad r^\star = \min \arg\min_{r \in C_4} \kappa(x_r), \qquad c(0) = 0.$$

Read as: Serialize each normalized orientation with key κ, select the lexicographically smallest key, and break a tie by choosing the smallest rotation index.

Simple example. If the four byte keys are $[7, 3, 9, 4]$ in lexicographic order, select the second candidate. If two minima tie at indices 1 and 3, choose index 1.

Relationship and why the paper needs it. F74 turns the orbit into one deterministic image and makes the map measurable because only four candidates are compared. The special zero-image rule handles empty support.

F75. Translation invariance of normalization

$$N\{(t, 0) \cdot z\} = N(z) \qquad \text{for every } t \in \mathbb{Z}^2.$$

Read as: Translating an image before anchoring its bounding box does not change the anchored result.

Simple example. Whether an L-shape starts at $(0, 0)$ or $(10, -4)$, both become the same origin-anchored crop.

Relationship and why the paper needs it. This establishes invariance to the infinite translation subgroup and is the first step in proving Theorem 6.1.

F76. Transforming a moved image before testing rotations

$$y = (t, s) \cdot x, \qquad (0, r)(t, s) = (R_r t, r + s).$$

Read as: If y is a moved version of x, first rotating y by r is equivalent to a transformed translation $R_r t$ and the combined rotation $r + s$ applied to x.

Simple example. A 90° test rotation changes the direction of the pre-existing translation vector and adds one to the original quarter-turn index.

Relationship and why the paper needs it. This is F70 applied to the invariance proof. After normalization, the changed translation disappears and only a permutation of rotation candidates remains.

F77. Equality of normalized candidate sets

$$\begin{aligned} N\{(0, r) \cdot y\} &= N\{(0, r)(t, s) \cdot x\} \\ &= N\{(R_r t, r + s) \cdot x\} \\ &= N\{(0, r + s) \cdot x\}. \end{aligned}$$

Read as: Each normalized rotation candidate of y equals one normalized rotation candidate of x, with its index shifted by s.

Simple example. If $s = 1$, the list $(0, 1, 2, 3)$ is merely relabeled $(1, 2, 3, 0)$; its lexicographic minimum is unchanged.

Relationship and why the paper needs it. The first two lines use F76; the last uses translation invariance F75. Therefore F74 selects the same canonical image from both lists.

F78. Invariance of the canonicalizer

$$c(y) = c(x).$$

Read as: Any two images differing by an allowed lattice rigid motion have the same canonical representative.

Simple example. A shifted 270° L-shape and the origin-anchored L-shape yield identical c.

Relationship and why the paper needs it. F77 shows their candidate sets agree. F78 proves the invariance half of maximal invariance; the reverse half begins with F79.

F79. Equality to a common canonical image

$$c(x) = c(y) = z, \qquad z = (u, r) \cdot x, \qquad z = (v, s) \cdot y.$$

Read as: If two images canonicalize to the same z, then each can be moved to z by one of the transformations used during canonicalization.

Simple example. One image may need a shift and 90° turn, while the other needs a different shift and 180° turn; both reach the same canonical bitmap.

Relationship and why the paper needs it. This is the starting point for proving that equal canonical outputs cannot merge different group orbits.

F80. Recovering the transformation from x to y

$$y = (v, s)^{-1} \cdot z = (v, s)^{-1}(u, r) \cdot x.$$

Read as: Undo the transformation that takes y to z, then substitute the transformation that takes x to z; the composition maps x to y.

Simple example. If x is rotated to the canonical orientation and y is shifted to it, undoing the y motion after the x motion directly relates the two images.

Relationship and why the paper needs it. Closure and inverses from F70 ensure the composed motion remains in G_{rig}. Thus equality of canonical outputs implies orbit equivalence.

F81. Maximality of the lattice canonicalizer

$$c(x) = c(y) \quad \Longleftrightarrow \quad y = g \cdot x \text{ for some } g \in G_{\mathrm{rig}}.$$

Read as: Two lattice images have the same canonical representative exactly when they differ by an integer translation and quarter turn.

Simple example. All allowed moved copies of one L-shape agree under c, but a square cannot share that canonical image.

Relationship and why the paper needs it. F78 proves one direction and F79–F80 prove the other. F81 is the concrete instance of the abstract maximal-invariant definition F11.

F82. Two-site well space and nuisance group

$$\mathcal{W} = \mathcal{X}_{q,L}^{;2}, \qquad w = (x_1, x_2), \qquad G_{\mathrm{well}} = G_{\mathrm{rig}} \times G_{\mathrm{rig}}.$$

Read as: A well contains two labeled q-channel lattice images, and the declared nuisance model lets each site have its own rigid motion.

Simple example. Site 1 and site 2 form one outcome but receive different shifts and quarter turns.

Relationship and why the paper needs it. This explicitly chooses the product-action case F49, matching the imposed RxRx1 stress-test generator. It would be the wrong group for one shared motion.

F83. Well-level canonicalization (paper Eq. 6.3)

$$M_{\mathrm{well}}(x_1, x_2) = \big(c(x_1), c(x_2)\big), \qquad \mathcal{W}_{\mathrm{can}} = M_{\mathrm{well}}(\mathcal{W}).$$

Read as: Canonicalize the two labeled sites separately; the set of all resulting pairs is the well quotient space.

Simple example. A well becomes an ordered pair consisting of its canonical site-1 and site-2 multichannel crops.

Relationship and why the paper needs it. F81 gives a maximal invariant at one site, and Proposition 4.1 with the product action proves F83 is maximal for the well group in F82.

3.10 Section 6.2: characteristic quotient kernel

F84. Injective Euclidean encoding

$$\psi : \mathcal{W}_{\text{can}} \longrightarrow \mathbb{R}^{2qL^2}.$$

Read as: Place both canonical q-channel crops on fixed $L \times L$ canvases and serialize all $2qL^2$ pixel values into one vector.

Simple example. With two sites, six channels, and $L = 64$, the vector has $2 \cdot 6 \cdot 64^2 = 49{,}152$ entries.

Relationship and why the paper needs it. Injectivity means no canonical image information is lost during vectorization. It permits use of a standard Gaussian kernel while preserving distinction among quotient laws.

F85. Gaussian quotient kernel (paper Eq. 6.4)

$$k_\sigma(w, w') = \exp\left[-\frac{\|\psi\{M_{\text{well}}(w)\} - \psi\{M_{\text{well}}(w')\}\|_2^2}{2\sigma^2}\right].$$

Read as: Canonicalize both wells, vectorize them, compute squared Euclidean distance, and convert that distance into a similarity between zero and one.

Simple example. Identical quotient wells have distance zero and kernel value one. A vector distance equal to σ gives similarity $e^{-1/2} \approx 0.607$.

Relationship and why the paper needs it. F83 removes the declared nuisance, F84 retains the complete canonical outcome, and F85 supplies the kernel used by MMD and the randomization test.

F86. Ambient Gaussian kernel

$$k_G(u, v) = \exp\left\{-\frac{\|u - v\|_2^2}{2\sigma^2}\right\}, \qquad u, v \in \mathbb{R}^{2qL^2}.$$

Read as: This is the usual Gaussian radial-basis kernel on the serialized Euclidean vectors.

Simple example. Vectors one bandwidth apart have similarity $e^{-1/2}$.

Relationship and why the paper needs it. F85 is the pullback of F86 through $\psi \circ M_{\text{well}}$. Positive semidefiniteness and characteristicness therefore transfer to the quotient construction.

F87. Invariance in both kernel arguments

$$\begin{aligned} &M_{\text{well}}(g \cdot w) = M_{\text{well}}(w), \\ &k_\sigma(g \cdot w, w') = k_\sigma(w, w'), \qquad k_\sigma(w, g \cdot w') = k_\sigma(w, w'). \end{aligned}$$

Read as: Transforming either input well by an allowed site-specific motion cannot change its kernel similarity to another well.

Simple example. Rotate and shift either well before computing the kernel; canonicalization removes the change.

Relationship and why the paper needs it. The first formula follows from F83 and maximal invariance. Substituting it into F85 proves the second line, ensuring acquisition geometry cannot alter the test statistic under the exact action.

F88. Characteristicness through pushforward

$$\psi_{\#}P = \psi_{\#}Q \quad \Longrightarrow \quad P = Q.$$

Read as: If the serialized canonical vectors have the same distribution, then the original quotient-valued outcomes have the same distribution because ψ is injective with a measurable inverse on its image.

Simple example. Two distributions on canonical images cannot become indistinguishable merely because of vectorization; every vector maps back to its unique canonical image.

Relationship and why the paper needs it. The Gaussian kernel distinguishes all Euclidean laws. F88 pulls that property back to quotient laws, so zero population MMD means equality of the full quotient distributions, not only equality of selected moments.

The paper's finite directional Euler vector is a secondary endpoint computed after M_{well}. It is therefore exactly invariant under the declared action, but it is not injective and is neither claimed nor used as a maximal invariant. This distinction is why the canonical pixel endpoint is the sole primary test.

3.11 Section 7: exact design-based testing

F89. Quotient outcomes and paired assignment labels

$$q_{b,0}, q_{b,1} \in \mathcal{W}_{\text{can}}, \qquad Z_b \in \{0,1\}, \qquad Z = (Z_1, \ldots, Z_B).$$

Read as: In block b, order the two wells before looking at assignments; $q_{b,0}$ and $q_{b,1}$ are their quotient outcomes, and Z_b records whether the first well received condition 1.

Simple example. If $Z_b = 1$, the first well is treated and the second is control; if $Z_b = 0$, the roles are swapped.

Relationship and why the paper needs it. Deterministic ordering makes assignment labels well-defined. The randomization distribution varies Z while treating the potential outcomes as fixed under the sharp null.

F90. Complete paired assignment space

$$\Omega = \{0,1\}^B, \qquad Z \text{ is uniform on } \Omega.$$

Read as: Each of the B pairs has two possible labelings, giving 2^B equally likely complete assignments.

Simple example. With $B = 3$, there are $2^3 = 8$ swap patterns, each with probability $1/8$.

Relationship and why the paper needs it. This is the design assumption that makes complete enumeration exact. Metadata showing paired eligibility is not by itself proof of uniformity; the paper states this limitation.

F91. Treatment and control samples under a proposed assignment

$$Q_1(z) = \{q_{b,1-z_b} : b = 1, \dots, B\},$$
$$Q_0(z) = \{q_{b,z_b} : b = 1, \dots, B\}.$$

Read as: Under label vector z, choose from each ordered pair the well assigned to condition 1 and the complementary well assigned to condition 0.

Simple example. If $z_b = 1$, $1 - z_b = 0$, so the first ordered well $q_{b,0}$ enters condition 1 and the second $q_{b,1}$ enters condition 0, consistent with the paper's indexing convention.

Relationship and why the paper needs it. F89 defines the ordered outcomes and label; F91 constructs the two samples for every possible swap pattern used in F96.

F92. Population squared MMD (paper Eq. 7.1)

$$\mathrm{MMD}_k^2(P_1, P_0) = \mathbb{E}k(U, U') + \mathbb{E}k(V, V') - 2\mathbb{E}k(U, V),$$

Read as: Here $U, U' \overset{\text{iid}}{\sim} P_1$, $V, V' \overset{\text{iid}}{\sim} P_0$, and all four variables are mutually independent. Compare average similarity within treatment, average similarity within control, and twice the average similarity across treatment and control.

Simple example. If within-group kernel similarities are near one but cross-group similarity is small, MMD is large, indicating different quotient distributions.

Relationship and why the paper needs it. F59 defined MMD as a feature-mean distance. F92 is its kernel-expectation expansion, which leads to the computable empirical statistic F93.

F93. Empirical two-sample discrepancy (paper Eq. 7.2)

$$\begin{aligned}\widehat{\mathrm{MMD}}_u^2(z) = {} & \frac{1}{B(B-1)} \sum_{b \neq b'} k(q_{b,1-z_b}, q_{b',1-z_{b'}}) \\ & + \frac{1}{B(B-1)} \sum_{b \neq b'} k(q_{b,z_b}, q_{b',z_{b'}}) \\ & - \frac{2}{B^2} \sum_{b,b'} k(q_{b,z_b}, q_{b',1-z_{b'}}).\end{aligned}$$

Read as: Average all off-diagonal treatment-treatment similarities, all off-diagonal control-control similarities, and subtract twice the average of all control-treatment similarities.

Simple example. With $B = 2$, each within-condition term averages the two ordered off-diagonal kernel values; the cross term averages all four control-treatment comparisons.

Relationship and why the paper needs it. F93 estimates the structure in F92 and supplies one deterministic score for each assignment z. Because the observations are paired, the paper carefully calls it a prespecified discrepancy statistic rather than relying on the usual independent-sample unbiasedness interpretation.

F94. Label-invariant pooled set

$$Q_{\text{pool}} = \{q_{b,j} : b = 1, \ldots, B,\ j \in \{0, 1\}\}.$$

Read as: Pool both wells from every block without using their condition labels.

Simple example. Eight blocks contribute sixteen quotient outcomes to the pooled set.

Relationship and why the paper needs it. The bandwidth is selected from F94 so it remains exactly unchanged under all paired label swaps, a requirement for an exact randomization distribution.

F95. Prespecified Gaussian bandwidth

$$\sigma^2 = \frac{1}{2}\,\text{median}\left\{\|\psi(q) - \psi(q')\|_2^2 : q, q' \in Q_{\text{pool}},\ q \neq q',\ \|\psi(q) - \psi(q')\|_2 > 0\right\}.$$

Read as: Compute every nonzero squared distance between distinct pooled canonical outcomes, take the median, and divide by two to obtain σ^2.

Simple example. If the median nonzero squared distance is 18, set $\sigma^2 = 9$ and $\sigma = 3$.

Relationship and why the paper needs it. F95 tunes F85 without looking at labels. Because F94 is swap-invariant, the kernel is fixed throughout enumeration and does not adapt separately to favorable assignments.

F96. Test statistic and exact upper-tail p-value (paper Eq. 7.3)

$$S : \Omega \to \mathbb{R}, \qquad S(z) = \widehat{\text{MMD}}_u^2(z),$$
$$p(Z) = 2^{-B} \sum_{z \in \Omega} \mathbf{1}\{S(z) \geq S(Z)\}.$$

Read as: Evaluate the statistic for every possible paired assignment and report the fraction whose value is at least as large as the observed one.

Simple example. With $B = 3$, if one of eight assignments is at least as extreme as observed, $p = 1/8 = 0.125$.

Relationship and why the paper needs it. F90 gives equal weights 2^{-B}, F91 constructs each relabeling, and F93 computes its score. Exactness depends on the assignment mechanism and sharp null, not on asymptotic approximations or unbiased MMD estimation.

F97. Finite-sample validity statement

$$\mathbb{P}\{p(Z) \leq \alpha \mid \text{design information and sharp null}\} \leq \alpha, \qquad \alpha \in [0, 1].$$

Read as: Under the stated paired-randomization design and quotient sharp null, the chance of rejecting at level α is no greater than α.

Simple example. At $\alpha = 0.05$, a valid test falsely rejects at most 5% conditionally, although discreteness may make it conservative.

Relationship and why the paper needs it. F97 is the causal inferential guarantee. It applies to the invariant potential outcome sharp null, with well-defined treatments, no interference, swap-invariant retention, and uniform assignment.

F98. Rank-count proof of validity

$$N = 2^B, \qquad r(z) = \#\{u \in \Omega : S(u) \geq S(z)\}, \qquad p(z) = \frac{r(z)}{N},$$
$$m = \lfloor \alpha N \rfloor, \qquad \#\{z \in \Omega : r(z) \leq m\} \leq m.$$

Read as: $r(z)$ is the upper-tail rank count of assignment z, and the exact p-value is that count divided by the total number of assignments. With $m = \lfloor \alpha N \rfloor$, at most m assignments can have rank count at most m.

Simple example. With $B = 8$, $N = 256$. The reported primary $p = 0.0078125$ equals $2/256$, meaning two assignments were at least as extreme.

Relationship and why the paper needs it. At most m assignments can have rank count at most m; ties only enlarge the count. Taking $m = \lfloor \alpha N \rfloor$ proves F97 by elementary counting.

F99. Simulation effect strengths and enumeration size

$$\eta \in \{0, 0.35, 0.70, 1.00\}, \qquad 2^8 = 256 \text{ paired assignments per replicate.}$$

Read as: The simulations test no intrinsic effect and three increasing annulus effects, with eight pairs and complete enumeration in every replicate.

Simple example. At $\eta = 0$, each simulated unit's two treatment potential outcomes are bytewise identical; any systematic rejection caused only by acquisition is invalid.

Relationship and why the paper needs it. F99 operationalizes F96. The null setting checks type-I error, while positive η assesses ability of invariant endpoints to detect a genuine morphological change.

3.12 Sections 8–9: how the formulas produce the reported evidence

These sections contain numerical results rather than new general equations, but their interpretation depends directly on the formulas above.

F100. Simulation rejection fraction

$$\widehat{\rho}_\eta = \frac{\#\{\text{replicates with } p \leq 0.05\}}{250}.$$

Read as: At effect strength η, divide the number of rejected simulation replicates by 250.

Simple example. For the exact quotient at $\eta = 0$, the reported fraction 0.052 corresponds to 13/250 rejections and has exact Clopper–Pearson interval $(0.028, 0.087)$ after rounding.

Relationship and why the paper needs it. This empirical fraction checks F97 across independent simulated experiments. Near 0.05 at $\eta = 0$ supports calibration; its rise to 0.992 at $\eta = 1$ shows power for the simulated effect.

F101. Primary RxRx1 exact p-value resolution

$$B = 8, \qquad |\Omega| = 2^8 = 256, \qquad p = 0.0078125 = \frac{2}{256}.$$

Read as: The primary canonical-quotient statistic is at least matched by only two of the 256 paired assignments.

Simple example. The exact p-value grid has spacing $1/256 \approx 0.003906$; 0.0078 is the second grid point.

Relationship and why the paper needs it. This is F96 evaluated on the primary RxRx1 contrast. It is a finite-design statement conditional on the claimed assignment mechanism, not an asymptotic Gaussian approximation.

F102. Exact-invariance audit

$$\max \|\Delta \text{ canonical features}\| = 0, \qquad \max |\Delta \widehat{\mathrm{MMD}}_u^2| = 0, \qquad \max |\Delta p| = 0.$$

Read as: Across the audited exact lattice transformations, canonical features, the test statistic, and the exact p-value were unchanged to the stored numerical precision.

Simple example. Moving or quarter-turning any frozen well before canonicalization returned byte-identical quotient features.

Relationship and why the paper needs it. These checks are the software-level realization of F78, F83, and F87. They verify the implementation on generated transformations, not the universal theorem by themselves.

F103. Approximate-action diagnostic changes

$$\text{relative feature error}, \qquad |\Delta \widehat{\mathrm{MMD}}_u^2|, \qquad |\Delta p|.$$

Read as: For transformations outside the exact group, separately record representation change, statistic change, and p-value change relative to the exact-transform baseline.

Simple example. Under a 5° rotation, quotient pixels had relative feature error 0.8432 but $|\Delta p| = 0$ in the frozen primary contrast.

Relationship and why the paper needs it. F64 would give a population guarantee only if F57 and F63 were verified on the relevant class. The paper correctly treats Table 5 as empirical sensitivity evidence, not as an application of the theorem to a discontinuous canonicalizer.

3.13 Appendix A.1: detailed factorization proof

F104. Preimage of an arbitrary target event

$$E := h^{-1}(B), \qquad B \subseteq \mathcal{T} \text{ Borel}.$$

Read as: Let E be the set of original objects whose invariant target value lies in the chosen measurable set B.

Simple example. If h is pixel count and $B = \{3\}$, then E is the set of all three-pixel images.

Relationship and why the paper needs it. To prove the factorization F12, the appendix shows that every such event E is determined by M; equivalently, h is measurable with respect to $\sigma(M)$.

F105. Saturation of the event by maximal-invariant fibres

$$M^{-1}\{M(E)\} = E.$$

Read as: If an object shares its maximal-invariant value with any object in E, then it is itself in E.

Simple example. Every moved copy of a three-pixel image also has three pixels, so the complete orbit lies inside E.

Relationship and why the paper needs it. The easy inclusion is $E \subseteq M^{-1}\{M(E)\}$. The reverse uses maximality of M and invariance of h, expanded in F106.

F106. Invariance establishes the reverse saturation inclusion

$$y \in M^{-1}\{M(E)\} \Longrightarrow \exists z \in E : M(y) = M(z) \Longrightarrow y = g \cdot z,$$
$$h(y) = h(g \cdot z) = h(z) \in B.$$

Read as: Sharing a maximal-invariant value puts y in the same orbit as some $z \in E$; invariance then gives y the same target value, so $y \in E$.

Simple example. If z is a three-pixel L-shape and y is its rotated copy, both have pixel count three.

Relationship and why the paper needs it. This proves the nontrivial inclusion in F105. It is the set-level version of the statement that h is constant on fibres of M.

F107. Disjoint images of the event and its complement

$$M(E) \cap M(\mathcal{Y} \setminus E) = \varnothing.$$

Read as: No quotient label can be attained by both an object whose target lies in B and one whose target lies outside B.

Simple example. A single orbit label cannot simultaneously represent a three-pixel image and a non-three-pixel image when pixel count is invariant.

Relationship and why the paper needs it. If an s were in both images, F108 would place two same-orbit objects on opposite sides of E, contradicting invariance.

F108. Contradiction behind disjointness

$$M(y) = s = M(z), \qquad y \in E, \quad z \notin E, \qquad h(y) = h(z).$$

Read as: If the same quotient label came from both sides, maximality would make the two objects orbit-equivalent and invariance would force equal target values.

Simple example. A rotated copy cannot leave the event "pixel count equals three."

Relationship and why the paper needs it. Because $h(y) \in B$ and $h(z) \notin B$, the final equality is impossible. Hence F107 holds.

F109. Lusin separation of analytic images

$$M(E) \subseteq D_B, \qquad M(\mathcal{Y} \setminus E) \subseteq \mathcal{S} \setminus D_B, \qquad D_B \subseteq \mathcal{S} \text{ Borel}.$$

Read as: Although the two image sets need not themselves be Borel, a Borel set D_B can be placed between them because they are disjoint analytic sets.

Simple example. Think of D_B as the measurable collection of orbit labels assigned to the target event B.

Relationship and why the paper needs it. F107 provides disjointness; standard-Borel assumptions permit Lusin separation. This technical step is why the factor map $\bar{h}$ can be chosen Borel rather than merely set-theoretic.

F110. Recovering the original event from quotient labels

$$E = M^{-1}(D_B).$$

Read as: An object belongs to the target event exactly when its maximal-invariant label lies in the corresponding Borel set D_B.

Simple example. An image has three pixels exactly when its canonical-image label belongs to the set of three-pixel canonical images.

Relationship and why the paper needs it. F109 ensures all labels from E are inside D_B and all labels from the complement are outside. F110 proves that E is measurable using only M.

F111. Measurability with respect to the maximal invariant

$$h^{-1}(B) = E = M^{-1}(D_B) \in \sigma(M).$$

Read as: Every measurable event about h is the preimage of a measurable quotient event, so knowing M is enough to determine h measurably.

Simple example. The event "pixel count equals three" can be decided from the canonical representative alone.

Relationship and why the paper needs it. Because this holds for every Borel B, h is $\sigma(M)$-measurable. The measurable factorization lemma then produces F112.

F112. Existence of the factor map

$$\bar{h} : \mathcal{S} \to \mathcal{T}, \qquad h = \bar{h} \circ M.$$

Read as: There is a measurable rule on quotient labels whose composition with M reproduces the invariant target.

Simple example. $\bar{h}$ counts bright pixels in a canonical image.

Relationship and why the paper needs it. F111 supplies the hypothesis for the measurable factorization lemma. F112 is the detailed proof of Theorem 2.4's main claim F12.

F113. Uniqueness on the attained quotient range

$$\begin{gathered} h = \bar{h}_1 \circ M = \bar{h}_2 \circ M, \\ \bar{h}_1(s) = \bar{h}_1\{M(y)\} = h(y) = \bar{h}_2\{M(y)\} = \bar{h}_2(s), \qquad s \in M(\mathcal{Y}). \end{gathered}$$

Read as: If two factor maps reproduce h, choose any object with quotient label s; both maps must return its same h value.

Simple example. Two purported pixel-count rules on actual canonical images cannot disagree and still compose with M to give pixel count.

Relationship and why the paper needs it. The factor map is unique where quotient labels can occur. Values outside $M(\mathcal{Y})$ are irrelevant and need not be unique.

3.14 Appendix A.2–A.3: existence and measurability constructions

F114. Orbit-valued maximal invariant for compact actions

$$y \longmapsto G \cdot y.$$

Read as: Map an object to its entire orbit, viewed as a compact subset when a compact metrizable group acts continuously on a Polish space.

Simple example. Under a finite rotation group, return the finite set of all rotated versions of the image.

Relationship and why the paper needs it. This is an abstract existence construction for maximal invariants. The paper's lattice method is more explicit because translations are noncompact.

F115. Finite-group canonical representative

$$\begin{aligned} H = \{h_1, \ldots, h_m\}, &\qquad \nu : \mathcal{Y} \to [0,1] \text{ Borel and injective}, \\ c_H(y) = h_{j^\star} \cdot y, &\qquad j^\star = \min \arg\min_j \nu(h_j \cdot y). \end{aligned}$$

Read as: Apply every element of a finite group, score the results with an injective measurable code, choose the smallest score, and break ties by the smallest index.

Simple example. Serialize the four rotations of an image and select the lexicographically smallest byte vector.

Relationship and why the paper needs it. F115 is the general finite-group version of the rotation step F74. The actual lattice construction first removes noncompact translations by F72 and then applies this finite minimization.

F116. Measurability of the Haar kernel integrand

$$(g, m) \longmapsto \mathbf{1}\{g \cdot s(m) \in B\}.$$

Read as: Given a group element and quotient label, indicate whether transforming its selected representative lands in event B.

Simple example. For four rotations and an event "faces upward," the indicator is one only for rotations producing that orientation.

Relationship and why the paper needs it. Joint measurability of the action and the section makes F116 measurable; integrating it over g proves that $m \mapsto K_m(B)$ in F34 is measurable.

4 How the formulas depend on one another

The paper's argument can be read as the following dependency chain:

1. F1–F10 show that unrestricted contamination leaves only orbit-invariant information uniformly recoverable.
2. F11–F12 construct a complete measurable container for all that observable information.
3. F13–F19 turn this container into a causal potential outcome and identify its full treatment-specific law under standard causal assumptions.
4. F20–F31 ask a different question and prove that quotienting is statistically lossless only when the within-orbit raw law is parameter-free.
5. F32–F47 show that conditional Haar contamination is one concrete way to obtain such a common law; it is not assumed by the observability or causal-identification results.
6. F48–F55 prevent an incorrect quotient choice: independent component motions call for a product action, while a shared motion requires a diagonal quotient that retains relative configuration.
7. F56–F67 bound errors under approximate contamination, but only under explicit Lipschitz assumptions that the lattice canonicalizer may fail globally.
8. F68–F88 build and justify an exact maximal invariant and characteristic kernel for the declared lattice product action.
9. F89–F99 convert the invariant kernel into a finite-sample paired randomization test.
10. F100–F103 interpret the simulation and RxRx1 evidence; F104–F116 supply the descriptive-set-theoretic proof details.

5 What the empirical results do and do not establish

5.1 What they support

- In the null simulation, the exact quotient rejection fraction was 0.052, close to the nominal 0.05, while raw pixels rejected in 1.000 of replicates. This demonstrates the danger of treatment- and outcome-dependent acquisition in the constructed mechanism.
- Quotient-test power increased from 0.140 at $\eta = 0.35$ to 0.848 at 0.70 and 0.992 at 1.00.
- In the frozen primary RxRx1 contrast, canonical quotient pixels gave $\widehat{\mathrm{MMD}}_u^2 = 0.0036$ and exact paired-swap $p = 0.0078$.
- Canonical features and test outputs were exactly unchanged across the audited integer shifts and quarter turns, consistent with the declared group action.

5.2 What they do not establish

- The method does not recover the original absolute position or orientation. It deliberately declares them unidentifiable nuisance coordinates.
- The primary RxRx1 result is a reagent-specific contrast, not automatically a gene-level causal mechanism.
- The biological images are real, but the informative acquisition transformations in the stress test are imposed. The experiment demonstrates behavior under that controlled nuisance mechanism rather than proving that RxRx1 naturally followed it.
- Exact randomization validity is conditional on the actual within-pair assignment being uniform, well-defined reagent versions, no cross-well interference, and swap-invariant retention.
- Theorem 5.1 does not automatically certify the off-grid rotation, cropping, or noise experiments because the lexicographic canonicalizer can be discontinuous and the required Lipschitz property was not verified on the relevant image class.
- A maximal invariant is lossless for invariant targets, but not necessarily sufficient for every parameter of the raw observed-data law. The paper explicitly proves and illustrates this distinction.
- The conditional-Haar result is a compact-group corollary. The implemented lattice group contains the noncompact translation group $\mathbb{Z}^2$, so the lattice method uses deterministic translation normalization and does not rely on a normalized Haar probability on translations.

6 Bottom line

Under unrestricted group contamination, only orbit-invariant targets are recoverable. Standard causal assumptions identify their treatment-specific quotient law; lossless reduction additionally requires a parameter-free within-orbit kernel. The lattice algorithm and paired-swap test instantiate these results for the declared RxRx1 action.